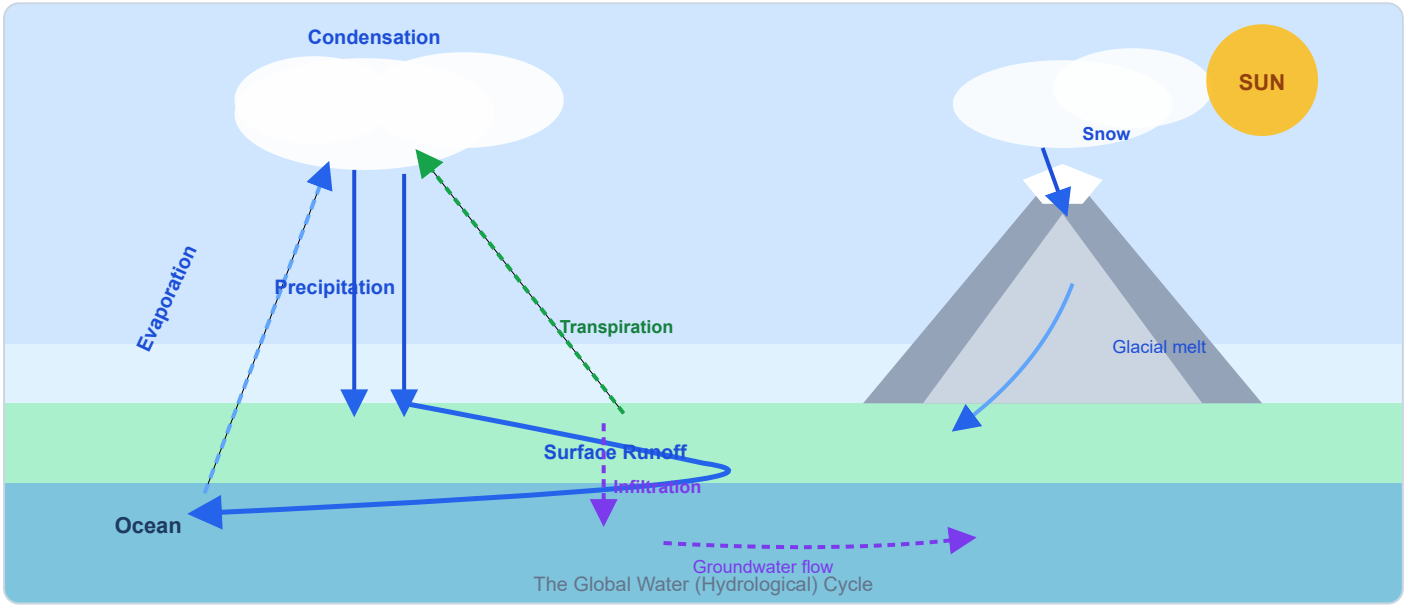
RAG Learning

Eduqas GCSE Geography (C/4) — Component 3

3.3 Water Management

1. The Global Water Cycle (Hydrological Cycle)

The water cycle describes the continuous movement of water through the Earth's systems. It has no start or end point — water is constantly being transferred between stores.



The Global Water (Hydrological) Cycle

Key Stores in the Water Cycle

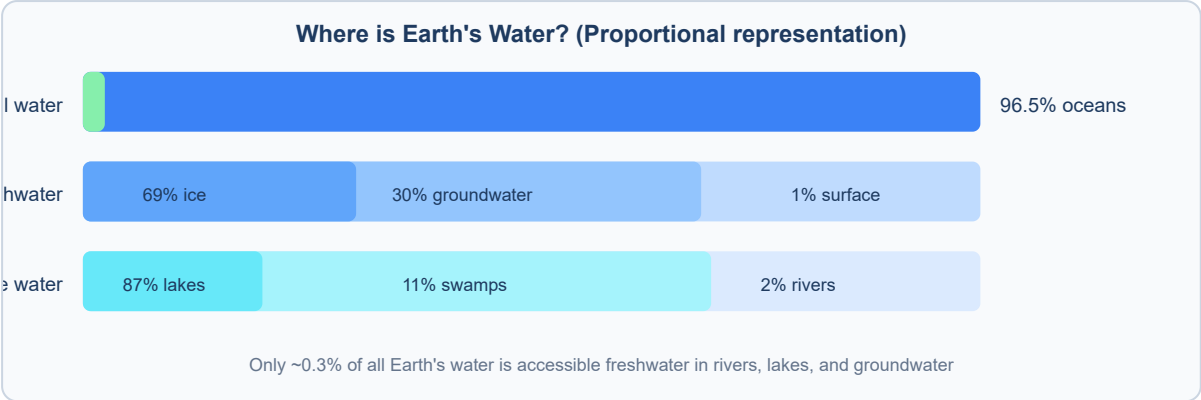
Store	% of total water	Notes
Oceans and seas	96.5% of all water	Saltwater — not directly usable for drinking/irrigation
Ice caps and glaciers	1.74% of all water (69% of freshwater)	Largest freshwater store, but mostly locked in Antarctica and Greenland

Groundwater	1.7% of all water (30% of freshwater)	Aquifers — vital freshwater store, slowly recharged
Surface water (rivers, lakes)	~0.013% of all water	Most easily accessible freshwater
Soil moisture	~0.001% of all water	Critical for agriculture and vegetation
Atmosphere	~0.001% of all water	Water vapour — drives weather systems
Biosphere (living organisms)	Tiny fraction	Important in nutrient cycling and transpiration

Key Processes

- **Evaporation** — liquid water converted to water vapour by solar energy (mainly from oceans).
- **Transpiration** — water released from plant leaves as vapour. Combined with evaporation = **evapotranspiration**.
- **Condensation** — water vapour cools, forms water droplets → clouds.
- **Precipitation** — water falls as rain, snow, hail, sleet.
- **Surface runoff** — water flows over land surface into rivers, streams, and eventually the sea.
- **Infiltration** — water soaks into the soil. Rate depends on soil type, vegetation, rainfall intensity.
- **Percolation** — water moves through soil into groundwater stores (aquifers).
- **Groundwater flow** — slow movement of water through rock and soil toward rivers and sea.

2. Global Water Distribution



Only **2.5% of all Earth's water is freshwater**, and of that, 69% is locked in ice caps and glaciers. Only around 0.3% of all water on Earth is accessible freshwater in rivers, lakes, and shallow groundwater — yet this tiny fraction must support all terrestrial life and human needs.

Global Distribution of Water Access

- **Water-rich regions:** tropical areas (Amazon basin, Congo basin — high rainfall year-round), temperate humid regions (NW Europe, E. North America, parts of East Asia).
- **Water-scarce regions:** Middle East and North Africa (MENA) — most water-scarce region globally; sub-Saharan Africa (economic scarcity); parts of Central/South Asia; W. USA and N. Mexico; Australia's interior.
- Access is affected not just by physical availability but also by infrastructure, governance, and wealth (economic water scarcity).

3. Water Demand

Global freshwater demand has increased **sixfold** over the past 100 years and continues to grow due to several drivers:

Driver	Detail
Population growth	World population reached 8 billion in 2022; UN projects 9.7 billion by 2050. More people directly means more water needed for drinking, cooking, sanitation, and food production.
Economic development	Industrialisation requires large amounts of water for manufacturing, cooling, and processing. Growing economies in Asia (India, China) rapidly increasing industrial water

	demand.
Rising living standards	Richer populations use more water: flush toilets, showers, washing machines, swimming pools, garden irrigation. Average domestic use: UK ~150 litres/day; Sub-Saharan Africa ~20 litres/day.
Changing diet	Meat production is extremely water-intensive: beef requires ~15,000 litres per kg (vs ~1,600 litres for wheat). As diets in NEEs shift toward more meat, agricultural water demand surges. China's meat consumption has doubled in 20 years.
Climate change	Higher temperatures increase evaporation, increasing agricultural irrigation needs. Changing precipitation patterns create more frequent droughts in already stressed regions.

Water Use by Sector

- **Agriculture:** ~70% of global freshwater withdrawals — mainly for irrigation. Responsible for 90% of consumptive water use (water that is not returned to the source).
- **Industry:** ~20% — cooling for power plants, manufacturing processes, mining.
- **Domestic:** ~10% — drinking, cooking, sanitation, hygiene.

4. Water Stress and Scarcity

Water stress — occurs when annual water withdrawals exceed 40% of the available renewable freshwater supply. Affects over 2 billion people globally.

Water scarcity — an absolute lack of supply. Two types:

- **Physical water scarcity** — genuinely insufficient rainfall or water sources. Middle East, N. Africa, parts of sub-Saharan Africa and Central Asia.
- **Economic water scarcity** — water is physically available but infrastructure and investment are absent to access it. Sub-Saharan Africa — water is present but wells, pipes, and treatment plants are not.

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Case Study: Day Zero — Cape Town, South Africa (2018)

Cape Town nearly became the first major city in the world to run out of municipal water supply. Following three years of below-average rainfall (2015–2017), reservoir levels fell to just 13.5% of capacity by January 2018. "Day Zero" — the day taps would be turned off and residents would have to queue at communal distribution points — was projected for April 2018.

- **Emergency measures:** Household water restricted to 50 litres per person per day (later 87 litres — WHO minimum is 50 litres). Agricultural water allocations cut. Hotels and restaurants imposed strict limits.
- **Outcome:** Day Zero was avoided — a combination of emergency restrictions, public compliance (90%+), minor rainfall improvement, and accelerated desalination and groundwater extraction. But only narrowly.
- **Lessons:** Urban water systems are vulnerable to climate variability; demand management (not just supply) is critical; groundwater over-extraction accelerated; desalination is expensive but a necessary insurance. Cape Town has since invested in diversifying water sources.

5. Impacts of Water Insecurity

Impact	Detail
Waterborne disease	Cholera, typhoid, dysentery, and diarrhoea spread through contaminated water. WHO: 1.8 million children under 5 die from diarrhoeal diseases each year — mostly from unsafe water. 2 billion people lack access to safe drinking water (2023).
Food insecurity	Agriculture is entirely dependent on water; water shortages → crop failures → famine. Sahel region (sub-Saharan Africa): recurring droughts causing chronic food insecurity for 300+ million people.
Conflict	Competition for scarce water resources fuels tensions. Israel/Palestine: Israeli settlements consume ~6x more water per capita than Palestinians; control of West Bank aquifers a major dispute. Water wars predicted as a defining feature of 21st-century geopolitics.

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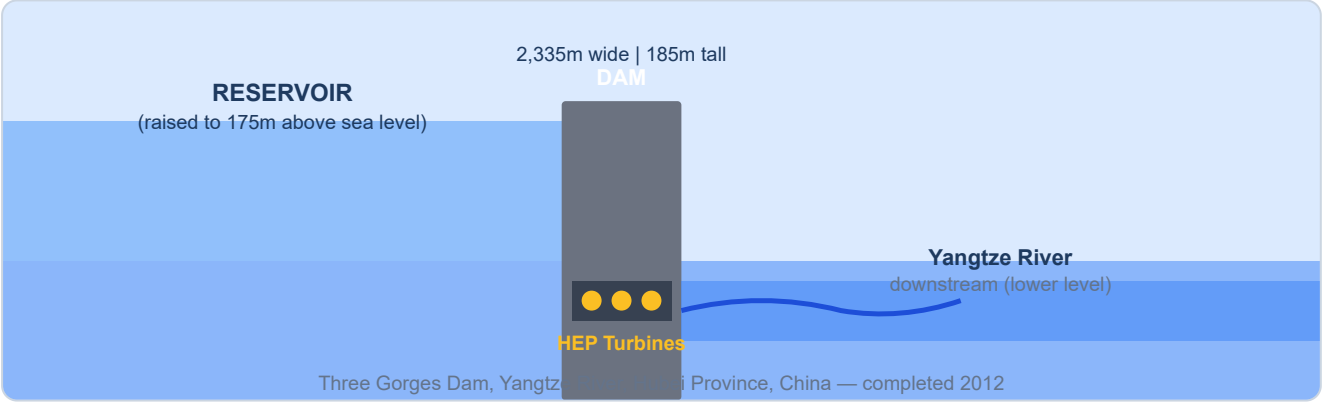
Economic underdevelopment	Without reliable water, industries cannot function, agriculture stagnates, and health costs drain resources. Women in LICs spend 200 million hours collectively per day collecting water — time unavailable for education or work.
Forced migration	Water scarcity drives people from their homes — "climate refugees." Sahel droughts contributing to migration to North Africa and Europe. Pacific island communities (Kiribati, Tuvalu) threatened by saltwater intrusion into freshwater supplies.
Gender inequality	In many LICs, women and girls are responsible for water collection — walking 2–6 hours/day. Girls miss school; women cannot participate in economic activity. Providing clean water to villages often dramatically improves girls' school attendance.

6. Large Dam Case Study: Three Gorges Dam, China



Case Study: Three Gorges Dam, China

Location: Yangtze River, Hubei Province, central China. Completed: 2012 (dam wall complete 2006; full capacity 2012). **Size:** 2,335m long, 185m tall — world's largest hydroelectric dam by power output. Cost: ~$37 billion.

Benefits

- **Hydroelectric power:** 22,500 megawatts (MW) capacity — generates ~3% of China's total electricity. Equivalent to replacing ~50 million tonnes of coal per year. Reduces air pollution and CO_2 emissions.

- **Flood control:** Before the dam, major floods devastated the Yangtze floodplain periodically (1954 flood killed 33,000; 1998 flood killed 4,000). The reservoir can store flood water, protecting ~15 million people and 1.5 million hectares of farmland downstream.
- **Navigation:** Raised water levels have allowed larger cargo ships to navigate 2,400km further inland to Chongqing (a city of 30 million). Shipping freight on the Yangtze increased fivefold since completion.
- **Water supply:** The reservoir stores water for dry-season use, supplying municipalities and agriculture downstream.
- **Economic development:** cheap electricity powered industrial development in western China; reduced coal dependency.

Costs and Controversies

- **Displacement:** 1.24 million people were forcibly relocated from the reservoir area (flooding 13 cities, 140 towns, 1,350 villages). Many were moved without adequate compensation or support. "Forced migrants" often relocated to marginal land; poverty increased for many families.
- **Cultural heritage loss:** Over 1,000 archaeological and historical sites submerged, including ancient temples, cliff carvings, and palaces. Many were inadequately documented before flooding.
- **Ecosystem impacts:** Chinese Sturgeon (critically endangered, ancient species) populations collapsed — spawning grounds flooded, migration blocked. Baiji river dolphin — declared functionally extinct 2006, partly due to dam effects. The Yangtze is now considered one of the world's most threatened river ecosystems.
- **Sediment trapping:** The dam traps sediment that would naturally enrich the Yangtze delta. Downstream erosion and wetland degradation have resulted. Effectiveness of flood control is also reducing as sediment fills the reservoir.
- **Landslide risk:** The reservoir's fluctuating water levels have destabilised slopes, triggering more than 90 landslides along the reservoir shoreline since filling. An estimated 530 potential landslide sites have been identified.
- **Induced seismicity:** The weight of water in the reservoir has been linked to increased minor earthquake frequency in the region.

Overall evaluation: The Three Gorges Dam is an engineering achievement that has brought real flood control and energy benefits to China. However, the human, ecological, and cultural costs are

immense and poorly compensated. It stands as a cautionary example for large dam projects globally.

Exam Tip: When evaluating the Three Gorges Dam, structure around economic, social, and environmental impacts — both positive and negative. Examiners want a balanced argument. A strong conclusion might be: "While the dam provides significant energy and flood control benefits for China's economy and population, the displacement of 1.24 million people and devastating ecological impacts on the Yangtze ecosystem raise serious questions about whether the project represented sustainable development."

7. Other Water Management Strategies

Water Transfers

Water transfer — moving water from a water-surplus area to a water-deficit area via canals, pipelines, or tunnels.

China's South-North Water Transfer Project — the world's largest water transfer scheme. Three planned routes move water from the Yangtze River basin (water surplus) northward to the Yellow River basin and Beijing/Tianjin (severe water deficit). Central Route (complete, 2014): 1,432km canal, carries 9.5 billion m^3/year to Beijing. Total eventual capacity: 44.8 billion m^3/year. Cost: ~$62 billion. Criticisms: further disrupts Yangtze ecosystem; displacement of 350,000+ people for the Central Route alone; may not solve underlying over-abstraction in the north.

Desalination

Desalination removes salt from seawater or brackish water to produce drinking water. Two main methods: reverse osmosis (forcing seawater through membranes) and thermal distillation (evaporating and condensing water).

- **Where used:** Saudi Arabia (world's largest capacity — ~21% of global desalination), UAE, Israel, Spain, USA (California), Australia.
- **Advantages:** provides water in arid areas with no alternative freshwater sources; technology is improving and costs falling.

- **Disadvantages:** extremely energy-intensive — Saudi Arabia uses oil to power desalination, contributing significantly to emissions. Cost: ~$1/m³ vs $0.05/m³ for groundwater extraction. Brine disposal — the concentrated salt residue is pumped back into the sea, harming marine ecosystems around discharge points.

Groundwater Extraction

Aquifers — underground layers of porous rock containing water — are an important freshwater source but can be over-exploited.

Ogallala Aquifer (USA) — one of the world's largest aquifers, underlying 8 states (Kansas, Nebraska, Texas, etc.), supporting the "breadbasket" of the USA. Being pumped out 8–10x faster than natural recharge rate. Water table has dropped 30% in 50 years. If exhausted, ~$20bn/year in agriculture would be unviable. A "slow-motion crisis" with no easy solution.

Rainwater Harvesting

Low-tech, low-cost collection of rainwater for later use. Appropriate technology suitable for LICs.

- **India's johads** — traditional earthen check dams and ponds that capture monsoon rainfall. In Rajasthan, NGO Tarun Bharat Sangh has restored 11,800 johads in 1,000 villages, bringing rivers back to life that had been dry for decades. Proven, low-cost, community-maintained.
- **Rooftop collection** — gutters and pipes channel roof water to storage tanks. Common in rural areas of India, Tanzania, Nepal. WaterAid installs these widely. Low maintenance, no electricity needed.

Greywater Recycling

Reusing household water from baths, showers, and washing machines for toilet flushing and garden irrigation — reducing demand for treated drinking water by 30–50% in domestic settings. Common in water-stressed areas of Australia, Israel, and parts of Europe.

8. Sustainable Water Management

Integrated Water Resource Management (IWRM) — a planning approach that coordinates the development and management of water, land, and related resources across all sectors (agriculture,

industry, domestic) and scales (local to national), aiming for efficiency, equity, and environmental sustainability.

Case Study: Singapore — "Four National Taps" Strategy

Singapore has no natural lakes or rivers and is entirely dependent on rainfall and water imported from Malaysia. Facing extreme vulnerability, Singapore developed a world-leading water security strategy:

1. **Imported water** (from Malaysia — water agreements ending 2061; Singapore is preparing for full independence).
2. **Local catchment water** — two-thirds of Singapore's land area drains into 17 reservoirs.
3. **NEWater** — highly purified reclaimed water from wastewater, treated to drinking water standard (and beyond — used in semiconductor manufacturing). Meets 40% of current needs; planned to meet 55% by 2060.
4. **Desalinated water** — 5 desalination plants, planned to meet 30% of needs by 2060.

Singapore now recycles 40% of its water and is one of the few countries in the world that has definitively solved its water security problem — a model for water-scarce megacities globally.

Case Study: Israel — World Leader in Water Efficiency

Israel receives an average of only 500mm rainfall per year (but ranges from 800mm in the north to 30mm in the Negev desert). Yet it is self-sufficient in food production and has a thriving agricultural export industry.

- **Drip irrigation** — invented in Israel in the 1960s (Netafim company). Delivers water directly to plant roots, reducing agricultural water use by 30–60% compared to traditional irrigation. Now used on 10% of irrigated land globally.
- **Treated wastewater reuse** — 85% of Israel's wastewater is treated and reused for agriculture — highest rate in the world. (Spain is 2nd at ~17%.) Virtually all irrigation in the Negev uses recycled water.
- **Desalination** — 5 large plants supply ~80% of Israel's drinking water. Sorek plant (largest reverse osmosis plant when built, 2013) produces water at just $0.54/m^3.

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- **National water grid** — National Water Carrier (1964) transfers water 130km from the Sea of Galilee to the drier south.

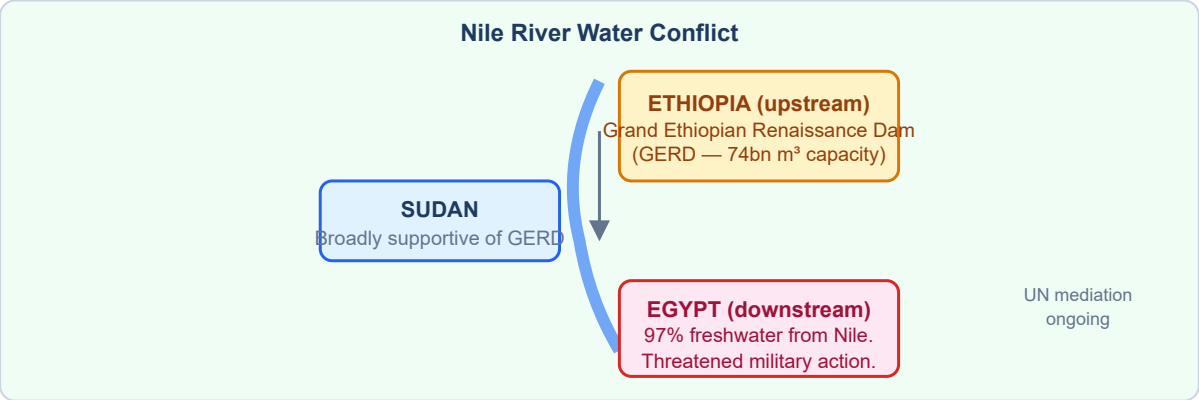
Appropriate Technology: WaterAid

WaterAid is a UK-based NGO providing safe water and sanitation to communities in LICs using appropriate, affordable technology:

- **Hand pumps** — manually operated borehole pumps (e.g. India Mark II). Durable, repairable by local communities with basic training. Costs ~$25 to provide one person with clean water access for life.
- **Biosand filters** — sand and gravel filter housed in concrete or plastic container. Removes 98%+ of bacteria, parasites, and turbidity from water. Easy to construct and maintain locally.
- **Pit latrines and CLTS** — Community-Led Total Sanitation reduces open defecation, which is a primary source of water contamination.

9. Water Conflicts

There are 310 transboundary river basins in the world — rivers shared between two or more countries. When water is scarce, sharing agreements become contentious and conflict can result.



River Nile Conflict — Ethiopia, Egypt, Sudan

Background: The Nile is shared by 11 countries but historically Egypt and Sudan had monopoly control over its waters under a 1959 Anglo-Egyptian agreement (Egypt: 55.5 billion m^3/year; Sudan: 18.5 billion m^3/year — other upstream nations excluded entirely).

- **GERD (Grand Ethiopian Renaissance Dam):** Ethiopia began construction in 2011 on the Blue Nile (source of 85% of Nile water). Capacity: 74 billion m^3 — Africa's largest hydroelectric dam. Expected to provide 15,000 MW, transforming Ethiopia's energy sector (60% currently lack electricity).
- **Egypt's position:** Egypt depends on the Nile for 97% of its freshwater. President Sisi stated: "No one can take a drop of Egypt's water." Egypt has threatened military action to protect Nile water rights. Fears GERD filling will reduce Nile flow for years.
- **Sudan's position:** Initially opposed (feared flooding), later became supportive as it benefits from GERD's flood control and cheap electricity.
- **Current status:** Ethiopia began filling GERD in 2020. AU and UN mediation ongoing with no binding agreement as of 2024. A key test of international water law (no universal treaty governing shared rivers).

Other Major Water Conflicts

- **Colorado River (USA/Mexico)** — heavily over-allocated (commitments exceed average flow by 20%+). Sometimes runs dry before reaching the sea in Mexico. Seven US states fight over allocations. 2023: Lake Mead (USA's largest reservoir) fell to 27% capacity.
- **Aral Sea (Central Asia)** — once the world's 4th largest lake (~68,000 km^2). Soviet cotton irrigation projects in the 1960s–80s diverted its feeder rivers (Amu Darya and Syr Darya). By 2007, the lake had shrunk to ~10% of its original size — one of the worst environmental disasters in history. Former fishing communities now live in a desert. Dust from the exposed lakebed causes health problems. Kazakhstan's North Aral Sea has been partially restored by a dam (2005).

10. Future Challenges

Water insecurity is expected to worsen significantly as a result of climate change, population growth, and economic development:

- **Climate change impacts on water:**
  - Precipitation is becoming more *variable* — intense floods and prolonged droughts increasing in frequency.

  - Glaciers that supply meltwater to major rivers (Ganges, Yangtze, Colorado) are retreating — short-term increase in meltwater, then long-term decline as glaciers disappear.
  - Sea-level rise causes saltwater intrusion into coastal aquifers (Bangladesh, Maldives, Florida) — contaminating freshwater supplies.
  - Higher temperatures increase evaporation, increasing irrigation demand while reducing supply.

- **5 billion people** could face water stress by 2050 (UN estimate).
- **Food security:** Feeding 10 billion people by 2050 will require 50% more food production — but water for agriculture is already overextended in many regions.

Solutions Require Multi-Level Action

Scale	Action needed
International	Transboundary water treaties; reform of water law; climate finance for water adaptation in LICs; support for SDG 6
National	Integrated water resource management; water pricing reform (currently heavily subsidised — distorting usage); infrastructure investment; drought contingency planning
Technology	More efficient irrigation (drip systems globally could save 30%+ of agricultural water); desalination innovation; wastewater recycling; atmospheric water generation
Behaviour	Reducing meat consumption; shorter showers; stopping water waste in industry and agriculture; consumer awareness of "virtual water" (water embedded in products)

SDG 6 (Sustainable Development Goal 6) — "Ensure availability and sustainable management of water and sanitation for all" by 2030. Progress is severely off-track: 2 billion people still lack safe drinking water; 3.6 billion lack safely managed sanitation. Would require a tenfold increase in investment to meet targets.

Virtual water — the water embedded in products during their production. 1kg of beef contains ~15,000 litres of virtual water; 1 cotton t-shirt contains ~2,700 litres. Trade in virtual water means rich

countries effectively "import" water from water-scarce countries when they buy agricultural products from those regions.

Exam Tip: For a "to what extent" question on water management, a strong structure is: argue one side (e.g. large-scale engineering solutions like dams are effective because...) → counter-argue (but they have significant costs because...) → consider alternative approaches (sustainable/appropriate technology is more equitable because...) → reach a justified conclusion. Refer to at least two case studies for 8-mark answers.

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